```latex
\documentclass[12pt]{article}

\usepackage{amsmath, amssymb, siunitx}

\usepackage[margin=1in]{geometry}

\title{NCHE 312 Physical Chemistry -- Class Test 5 \\ Marking Memo}

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\begin{document}


\maketitle


\section*{Question 1 (5 marks)}

Reaction: $2A + B \rightarrow 2C + 3D$ \\

Given rate law:

\[

\frac{d[C]}{dt} = k [A][B][C]

\]


\subsection*{(i) Reaction rate $v$}

\[

v = \frac{1}{\nu_C} \frac{d[C]}{dt} = \frac{1}{2} \frac{d[C]}{dt}

\]

\[

v = \frac{1}{2} k [A][B][C]

\]

\textbf{[3 marks]}


\subsection*{(ii) Units of $k$}
```

\[

\frac{d[C]}{dt} \text{ has units: } \si{\mol\per\cubic\deci\metre\per\second} = \si{\molar\per\second}

\]

\[

[A][B][C] \text{ has units: } \si{\molar^3}

\]

\[

[k] = \frac{\si{\molar\per\second}}{\si{\molar^3}} = \si{\molar^{-2}\per\second}

\]

In \si{\dm^3\per\mol} units:

\[

\boxed{\si{\dm^6\per\mol^2\per\second}}

\]

\textbf{[2 marks]}

\section*{Question 2 (5 marks)}

Reaction: $A + 3B \rightarrow C + 2D$ \\

Given:

\[

-\frac{d[B]}{dt} = \SI{2.7}{\mol\per\dm^3\per\second}

\]

\subsection*{(i) Reaction rate}

\[

v = -\frac{1}{3} \frac{d[B]}{dt} = \frac{1}{3} \times \SI{2.7}{\mol\per\dm^3\per\second} = \SI{0.9}{\mol\per\dm^3\per\second}

\]

\textbf{[2 marks]}

\subsection*{(ii) Rates of formation/consumption}

\[

-\frac{d[A]}{dt} = v = \SI{0.9}{\mol\per\dm^3\per\second}

\]

\[

\frac{d[C]}{dt} = v = \SI{0.9}{\mol\per\dm^3\per\second}

\]

\[

\frac{d[D]}{dt} = 2v = \SI{1.8}{\mol\per\dm^3\per\second}

\]

\textbf{[1 mark each]}

\[

\boxed{v = \SI{0.9}{\mol\per\dm^3\per\second},\quad -\frac{d[A]}{dt} = \SI{0.9}{\mol\per\dm^3\per\second},\quad \frac{d[C]}{dt} = \SI{0.9}{\mol\per\dm^3\per\second},\quad \frac{d[D]}{dt} = \SI{1.8}{\mol\per\dm^3\per\second}}

\]

\section*{Question 3 (5 marks)}

Reaction: $2A \rightarrow P$, second order \\

Integrated rate law:

\[

\frac{1}{[A]_t} - \frac{1}{[A]_0} = 2kt

\]

\[

k = \SI{4.3e-4}{\per\molar\per\second},\quad [A]_0 = \SI{0.21}{\molar},\quad [A]_t = \SI{0.01}{\molar}

\]

\[

\frac{1}{\SI{0.01}{\molar}} - \frac{1}{\SI{0.21}{\molar}} = \SI{100}{\per\molar} - \SI{4.7619}{\per\molar} = \SI{95.2381}{\per\molar}

\]

\textbf{[1 mark]}

\[

\SI{95.2381}{\per\molar} = 2 \times (\SI{4.3e-4}{\per\molar\per\second}) \times t

\]

\[

\SI{95.2381}{\per\molar} = \SI{8.6e-4}{\per\molar\per\second} \times t

\]

\[

t = \frac{\SI{95.2381}{\per\molar}}{\SI{8.6e-4}{\per\molar\per\second}} = \SI{110742}{\second}

\]

\textbf{[2 marks]}

Conversions:

\[

t = \frac{\SI{110742}{\second}}{60} = \SI{1845.7}{\minute}

\]

\[

t = \frac{\SI{1845.7}{\minute}}{60} = \SI{30.76}{\hour}

\]

\textbf{[2 marks]}

\[

\boxed{t = \SI{1.107e5}{\second} = \SI{1.846e3}{\minute} = \SI{30.8}{\hour}}

\]

\section*{Question 4 (5 marks)}

Arrhenius equation:

\[

\ln\frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)

\]

\[

k_1 = \SI{3.98}{\dm^3\per\mol\per\second},\quad T_1 = \SI{288}{K}

\]

\[

k_2 = \SI{14.3}{\dm^3\per\mol\per\second},\quad T_2 = \SI{299}{K}

\]

\[

R = \SI{8.314}{\J\per\mol\per\K}

\]

\[

\ln\left(\frac{\SI{14.3}{\dm^3\per\mol\per\second}}{\SI{3.98}{\dm^3\per\mol\per\second}}\right) = \ln(3.593) = 1.279

\]

\textbf{[1 mark]}

\[

\frac{1}{\SI{288}{K}} - \frac{1}{\SI{299}{K}} = \SI{0.0034722}{\per\K} - \SI{0.0033445}{\per\K} = \SI{0.0001277}{\per\K}

\]

\textbf{[1 mark]}

\[

1.279 = \frac{E_a}{\SI{8.314}{\J\per\mol\per\K}} \times \SI{0.0001277}{\per\K}

\]

\[

E_a = \frac{1.279 \times \SI{8.314}{\J\per\mol\per\K}}{\SI{0.0001277}{\per\K}} = \frac{\SI{10.633}{\J\per\mol}}{\SI{0.0001277}{\per\K}} = \SI{83260}{\J\per\mol}

\]

\textbf{[2 marks]}

\[

\boxed{E_a = \SI{83.3}{\kJ\per\mol}}

\]

\textbf{[1 mark]}

\section*{Question 5 (10 marks)}

Reaction: $2A \rightarrow B$ \\

Data:

\[

\begin{array}{c|cccccc}

t/\text{min} & 0 & 10 & 20 & 30 & 40 & \infty \\

\hline

[B]/\si{\molar} & 0 & 0.089 & 0.153 & 0.200 & 0.230 & 0.312 \\

\end{array}

\]

\subsection*{Step 1: Initial $[A]$}

\[

[A]_0 = 2 \times [B]_\infty = 2 \times \SI{0.312}{\molar} = \SI{0.624}{\molar}

\]

\textbf{[2 marks]}

\subsection*{Step 2: $[A]_t$}

\[

[A]_t = [A]_0 - 2[B]_t

\]

\[

\begin{array}{c|cc}

t/\text{min} & [B]/\si{\molar} & [A]/\si{\molar} \\

\hline

0 & 0 & 0.624 \\

10 & 0.089 & 0.446 \\

20 & 0.153 & 0.318 \\

30 & 0.200 & 0.224 \\

40 & 0.230 & 0.164 \\

\end{array}

\]

\textbf{[2 marks]}

\subsection*{Step 3: $1/[A]$ vs t}

\[

\begin{array}{c|c}

t/\text{min} & 1/[A]/\si{\per\molar} \\

\hline

0 & 1.602 \\

10 & 2.242 \\

20 & 3.145 \\

30 & 4.464 \\

40 & 6.098 \\

\end{array}

\]

\textbf{[2 marks]}

\subsection*{Step 4: Slope and k}

\[

\text{slope} = \frac{\SI{6.098}{\per\molar} - \SI{1.602}{\per\molar}}{\SI{40}{min} - \SI{0}{min}} = \frac{\SI{4.496}{\per\molar}}{\SI{40}{min}} = \SI{0.1124}{\per\molar\per\minute}

\]

For $2A \rightarrow P$: $\frac{1}{[A]_t} - \frac{1}{[A]_0} = 2kt$, so slope $= 2k$:

\[

k = \frac{\SI{0.1124}{\per\molar\per\minute}}{2} = \SI{0.0562}{\per\molar\per\minute}

\]

\textbf{[3 marks]}

\subsection*{Step 5: Order}

Linear $1/[A]$ vs t $\Rightarrow$ \textbf{second order}

\textbf{[1 mark]}

\[

\boxed{\text{Order} = 2,\quad k = \SI{5.62e-2}{\per\molar\per\minute}}

\]

\section*{Question 6 (10 marks)}

Electrode: Pt|H$_2$|H$^+$, $T = \SI{298}{K}$ \\

Data:

\[

\begin{array}{c|ccccc}

\eta/\si{\milli\volt} & 50 & 100 & 150 & 200 & 250 \\

\hline

j/\si{\milli\ampere\per\square\centi\metre} & 2.66 & 8.91 & 29.9 & 100 & 335 \\

\end{array}

\]

\subsection*{(i) Transfer coefficient α}

Tafel equation: $\eta = a + b \ln j$, $b = \frac{RT}{\alpha n F}$ \\

$n = 1$, $R = \SI{8.314}{\J\per\mol\per\K}$, $F = \SI{96485}{\C\per\mol}$ \\

b in mV:

\[

b = \frac{\SI{8.314}{\J\per\mol\per\K} \times \SI{298}{K}}{\alpha \times 1 \times \SI{96485}{\C\per\mol}} \times 1000 = \frac{\SI{25.7}{mV}}{\alpha}

\]

Compute $\ln j$:

\[

\begin{array}{c|c|c}

j/\si{mA\per\cm^2} & \ln(j/\si{mA\per\cm^2}) & \eta/\si{mV} \\

\hline

2.66 & 0.978 & 50 \\

8.91 & 2.187 & 100 \\

29.9 & 3.398 & 150 \\

100 & 4.605 & 200 \\

335 & 5.814 & 250 \\

\end{array}

\]

Slope:

\[

\frac{\SI{250}{mV} - \SI{50}{mV}}{5.814 - 0.978} = \frac{\SI{200}{mV}}{4.836} = \SI{41.36}{mV}

\]

Thus:

\[

\frac{\SI{25.7}{mV}}{\alpha} = \SI{41.36}{mV} \quad \Rightarrow \quad \alpha = \frac{25.7}{41.36} = 0.62

\]

\textbf{[3 marks]}

\textbf{Comment:} $\alpha \approx 0.62$ is typical for hydrogen evolution reaction, indicating an asymmetric energy barrier. \textbf{[2 marks]}

\subsection*{(ii) Exchange current density j_0}

From Tafel:

\[

\eta = \frac{RT}{\alpha F} \ln\frac{j}{j_0}

\]

Using $\eta = \SI{50}{mV} = \SI{0.050}{V}$, $j = \SI{2.66}{mA\per\cm^2}$:

\[

\SI{0.050}{V} = \frac{\SI{0.0257}{V}}{0.62} \ln\left(\frac{\SI{2.66}{mA\per\cm^2}}{j_0}\right)

\]

\[

\SI{0.050}{V} = \SI{0.04145}{V} \ln\left(\frac{\SI{2.66}{mA\per\cm^2}}{j_0}\right)

\]

\[

\ln\left(\frac{\SI{2.66}{mA\per\cm^2}}{j_0}\right) = \frac{\SI{0.050}{V}}{\SI{0.04145}{V}} = 1.206

\]

\[

\frac{\SI{2.66}{mA\per\cm^2}}{j_0} = e^{1.206} = 3.34

\]

\[

j_0 = \frac{\SI{2.66}{mA\per\cm^2}}{3.34} = \SI{0.796}{mA\per\cm^2}

\]

\textbf{[3 marks for calculation, 2 marks for final]}

\[

\boxed{\alpha = 0.62,\quad j_0 = \SI{0.80}{mA\per\cm^2}}

\]

\section*{Question 7 (10 marks)}

First-order decomposition in aqueous medium \\

Data:

\[

\begin{array}{c|cccc}

T/\si{\celsius} & 0 & 20 & 40 & 60 \\

\hline

k/\SI{e-5}{min^{-1}} & 2.46 & 43.5 & 576 & 5480 \\

\end{array}

```
\]

\subsection*{Step 1: Convert to Kelvin and to \si{\per\second}}

\[

\begin{array}{c|c|c|c|c}

T/\si{K} & 1/T/\SI{e-3}{K^{-1}} & k/\si{min^{-1}} & k/\si{s^{-1}} & \ln(k/\si{s^{-1}}) \\

\hline

273 & 3.663 & 2.46\times10^{-5} & 4.10\times10^{-7} & -14.71 \\

293 & 3.413 & 4.35\times10^{-4} & 7.25\times10^{-6} & -11.83 \\

313 & 3.195 & 5.76\times10^{-3} & 9.60\times10^{-5} & -9.25 \\

333 & 3.003 & 5.48\times10^{-2} & 9.13\times10^{-4} & -7.00 \\

\end{array}

\]

\textbf{[3 marks]}

\subsection*{Step 2: Slope from $\ln k$ vs $1/T$ (using T = 273 K and 333 K)}

\[

\text{slope} = \frac{-7.00 - (-14.71)}{(\SI{3.003e-3}{K^{-1}} - \SI{3.663e-3}{K^{-1}})} =
\frac{7.71}{-0.660\times10^{-3}\si{K^{-1}}} = \SI{-11681}{K}

\]

\textbf{[2 marks]}

\subsection*{Step 3: Activation energy}

\[

E_a = -\text{slope} \times R = \SI{11681}{K} \times \SI{8.314}{\J\per\mol\per\K} =
\SI{97100}{\J\per\mol} = \SI{97.1}{\kJ\per\mol}

\]

\textbf{[2 marks]}
```

\subsection*{Step 4: Pre-exponential factor A}

Using data at $T = \SI{273}{K}$:

\[

\ln k = \ln A - \frac{E_a}{RT}

\]

\[

-14.71 = \ln A - \frac{\SI{97100}{\J\per\mol}}{\SI{8.314}{\J\per\mol\per\K} \times \SI{273}{K}}

\]

\[

-14.71 = \ln A - \frac{\SI{97100}{\J\per\mol}}{\SI{2270}{\J\per\mol}}

\]

\[

-14.71 = \ln A - 42.78

\]

\[

\ln A = 28.07

\]

\[

A = e^{28.07} = \SI{1.55e12}{\per\second}

\]

\textbf{[3 marks]}

\[

\boxed{E_a = \SI{97.1}{\kJ\per\mol},\quad A = \SI{1.55e12}{\per\second}}

\]

\section*{Graph Notes for Students}

\begin{itemize}

\item \textbf{Question 5:} Plot $1/[A]$ (y-axis) vs t (x-axis). A straight line confirms second order. Slope $= 2k$.

\item \textbf{Question 6:} Plot η (mV) on y-axis vs $\ln j$ on x-axis. Slope $= b = RT/(\alpha F)$, intercept gives j_0.

\end{itemize}

\end{document}